Date: [Date]

To the Editorial Office, Remote Sensing Letters

Dear Editor,

We are pleased to submit our manuscript entitled “An Empirical Reproducibility Audit of MD-YOLO for Small-Object Detection in UAV Imagery” for consideration as a Research Letter in Remote Sensing Letters.

The manuscript reports an end-to-end empirical validation of the MD-YOLO detector for UAV aerial small-object detection on the real VisDrone2019-DET dataset (7 019 images, 381 963 bounding boxes). Executing the pipeline on real data—rather than relying on static source inspection—exposed four runtime defects, the most severe of which silently produced mAP = 0. After correction, we decompose the residual performance of the model into a resolved localization deficit and an open classification deficit attributable to a limited data–compute regime, and we further quantify a severe zero-shot domain shift on UAVDT-M (mean best-IoU 0.40 → 0.271). The work deliberately isolates three non-conflating failure categories and reports the negative results honestly, which we believe is of direct value to reproducible remote-sensing research.

We consider the contribution well suited to Remote Sensing Letters because it addresses a core UAV remote-sensing task—small-object detection in aerial imagery—and combines a cross-dataset domain-shift analysis (VisDrone → UAVDT-M) with a reproducibility methodology. The compact Letter format matches the focused scope of this study.

Authors: Qinmin Wang, Zhenzhen Hu, Hongyuan Wang, and Yuming Sun (corresponding author).

Manuscript type: Research Letter. Main text is within the 10-page limit; 1 figure and 3 tables (plus 1 supplementary table).

We confirm the following:

- This manuscript is original, has not been published previously, and is not under consideration by any other journal.
- We request subscription-based (non-open-access) publication; we do not wish to publish under an open-access licence and will not pay an article-processing charge.
- Full reproducibility scripts, pseudo-code, and supplementary diagnostic tables are supplied as Supplementary Material.
- All authors have approved the submission and declare no competing interests.

Thank you for your time and consideration. We look forward to your response.

Sincerely,

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